α-labeling of banana tree graphs

Borna Gojšić
Faculty of Electrical Engineering and Computing, Zagreb, Croatia
borna.gojsic@fer.hr

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A banana tree graph, denoted by $B(n,k)$, is a tree constructed by connecting n star graphs S_k to a root vertex. A graceful labeling assigns the numbers 0 through $|E|$ to vertices so that edge differences yield all values from 1 to $|E|$ exactly once; an α-labeling is a graceful labeling with an additional ordering property, motivated by work of Rosa [1], and is central in graph decomposition theory. This article builds upon previous research on graceful labelings of banana trees and explores their α-labelability. We utilize the symmetries of the $B(n,k)$ trees to give the number of all of the non-isomorphic α-labelings based on specific subsets of them which we call *proper* and *central* α-labelings. We classify all the α-labelings for all the $B(n,1)$, $B(1,k)$ and $B(2,k)$ trees. We present α-labelings of the $B(n,k)$ tree, for $k \geq \frac{n}{2}+1$ and $n \in \mathbb{N}$. Finally, we prove that the $B(n,k)$ tree is not α-labelable for $1 < k \leq \frac{n}{2}$ and any $n \geq 3$, classifying all the $B(n,k)$ trees based on their α-labelability.

References

[1] A. Rosa, On certain valuations of the vertices of a graph, in: Theory of graphs (Internat. Sympos., Rome, 1966), Gordon & Breach, New York, 1967, 349–355.